Jayadeep Kinavoor Madam

Platform Engineer | Certified Kubernetes Administrator | AWS Solutions Architect

AWS, Kubernetes, Terraform, Go, Prometheus, FluxCD, HashiCorp Vault, Github Actions, NixOS

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kmjayadeep

Vernier, Switzerland | Indian | 1995 September 15 | English (Fluent) | German (A1) | French (A1)

EXPERIENCE

SonarSource SA

Geneva, Switzerland
2024 Jan - Current

Platform Engineer (Full-time)

- Took ownership of Hashicorp Vault infrastructure that was previously managed by a third party company. Helped the team to gain expertise in managing vault, and convinced the management to bring vault management in-house. Ended up saving ~50K per year on support contract.
- Helped the team to adopt SRE principles and best practices to manage our services
- Undertook a massive migration project involving 10+ environments owned by our team in AWS. Implemented new VPCs to use centralized IPAM, GWLB for packet inspection and TGW for on-prem connectivity.
- Developed an SDK that enabled all teams to develop new self service actions in our IDP platform (port.io)
- Took end to end ownership of onboarding Gradle enterprise. Complete infra setup was done using Terraform, EKS and helm. It was used in 100+ repositories
- Undertook the hardening of 500+ IAM roles and policies in AWS to comply with ISO Audit requirements

Commercetools GmbH

Berlin, Germany
2022 Jan - 2023 Dec

Site Reliability Engineer (Full-time)

- Worked as Kubenetes SRE with focus on Observablity - Logging, Metrics and Alerts
- Took ownership and executed live migration of 7 production clusters in GKE
- Developed POC for scaling our prometheus setup using prometheus operator and sharding
- Gained expertise in setting up Horizontally scalable, Highly Available monitoring stack
- Contributed in setting up Meta-monitoring (monitoring the monitoring stack)
- On-call one day per week to take care of system upgrades, improvement of automation etc.

Dersalvador

Zurich (Remote)
2020 Jan - 2022 Jan

Senior Devops Engineer (Part-time, Weekends)

- Worked across multiple projects based on microservices
- Setup and maintained production grade kubernetes clusters using terraform in GCP
- Setup pipelines in Github actions, Gitlab CI, Flux and ArgoCD
- Designed and implemented event-driven asynchronous microservices pattern
- Developed REST APIs in NodeJS and GO
- Worked in various Containerization and Cloud Migration projects

Avesha Systems

Bangalore (Remote)
2019 Apr - 2021 Dec

Senior Software Engineer (Initially part-time, Transitioned to Full-time)

- Took responsibility of both development and operational roles based on requirements
- Designed and Developed Kubeslice, a multi cluster service mesh for Kubernetes in Golang using Operator SDK (Open-source)
- Designed and Developed Multi-cluster service discovery for Kubeslice
- Implemented DNS Server for Kubeslice using custom CoreDNS plugin
- Developed entire control plane components including CRDs, controllers, webhooks etc.
- Developed command line tool (golang) to interact with the mesh across different clusters
- Developed helm charts, deployment scripts, ArgoCD gitops setup and Jenkins pipelines
- Managed Nexus and Jenkins servers
- Developed Terraform scripts for deployment across cloud providers
- Developed custom E2e test framework for multi cluster testing

SAP Labs

Bangalore, India
2017 Jun - 2021 Jan

Full Stack Engineer (Full-time)

- Worked as Full stack developer across multiple teams (Java, Go and UI5)
- Responsible for end to end management of a cloud based full stack Application
- Developed jarvis, a tool to automate customer deployment which reduced application delivery time from 30 mins to 2 minutes
- Got selected for SAP Academy of Engineering, A prestigious training program for selected employees across the Globe

Sensomate Systems

Trivandrum, India
2016

Developer Intern (part time during studies)

- Worked as part of Development and debugging team
- Projects include BLE based Attendance management system (full stack), Municipality management app (ionic) and VPN application for Android

PERSONAL PROJECTS

Homelab Infrastructure

github.com/kmjayadeep/homelab-k8s

- Implemented GitOps automation using FluxCD with structured manifest organization and cluster-specific customizations
- Built Infrastructure-as-Code using Terraform, NixOS configurations, and Vault for secrets management
- Deployed production-grade networking with Cilium CNI, MetalLB load balancing, and automated certificate management
- Integrated sealed-secrets for secure GitOps workflows and established comprehensive monitoring stack with self-hosted applications
- Implemented self hosted Github runners using ARC for CI/CD pipelines

Majo Operator

github.com/kmjayadeep/majo-operator

- A Kubernetes Operator to automate periodic Database backups in Kubernetes
- Implemented using Operator SDK and GO
- Supports Mongodb and Mysql as Database sources, configurable as CRDs
- Supports a wide range of storage options like S3, Sftp, Ftp etc.

Watermark

github.com/kmjayadeep/watermark-microservices

- An example project which showcases asynchronous, event-driven cloud native solution to a problem
- Tech stack include kubernetes(GKE), terraform, pubsub, nodejs, mongodb, graphql, GRPC, KNative
- Includes one click script for provisioning, running tests and showing results

Wshare

github.com/kmjayadeep/wshare

- A simple, lightweight web app to share files through local & remote networks. Written in node.js using Express and ejs. Available in npm repository and github
- Supports drag and drop upload and download functionality

Micro Projects

resume: This resume is generated using golang and html templates
featureflag-nodejs: A feature flag implementation in nodejs
docker-compose-generator: Generate docker-compose.yaml from command line
nanodateformat: Lightweight date formatter for Javascript
digitalocean-k8s-terraform: A Terraform script to create Kubernetes cluster with ingress controller, cert-manager and metrics server

EDUCATION

International Institute of Information Techonology

IIIT, Bangalore
2021

PG Diploma in Cloud Computing

Kerala University

College Of Engineering, Trivandrum
2017

BTech in Computer Science - CGPA: 8.0

ACHIEVEMENTS & CERTIFICATIONS

- Certified Kubernetes Administrator - 95% (LF-kdin5ol6q2) - 2021
- Top contributor in open source project - "Litmus chaos engine" for Hactoberfest 2019
- Top contributor in open source project - "OpenEBS" for Hactoberfest 2018
- Front-end and Back-end certification from Freecodecamp
- Machine learning certificate from Coursera
- Outstanding student volunteer for IEEE in 2017